CHL5233: Exploratory Data Analysis

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This notebook follows the exploratory workflow in Telling Stories with Data, Chapter 11: describe individual variables, examine relevant relationships, and investigate what data are missing. It uses the processed country-year analysis dataset.

First look at the data

rows	countries	years	analysis_variables	duplicate_country_years
3720	186	2000–2019	16	0

iso	year	matmor	infmor	neomor	un5mor	gdp_1000	pop_dens	urban	age_dep	male_edu	temp	rainfall_1000	Drought	Earthquake	death_[illegible]	conflict
OMN	2006	18	10.7	5.6	12.5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	0	0	0
KHM	2003	404	61.8	29.7	78.0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	0	0	0
ZMB	2018	NA	43.9	24.7	65.4	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	0	0	0
IND	2011	197	43.0	30.6	55.0	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	1	1633	1
BLR	2019	NA	2.4	1.1	3.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	0	0	0
MRT	2017	766	52.6	33.1	77.7	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	1	0	0	0
RWA	2002	961	90.3	37.3	149.6	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	1	2044	1
SEN	2000	553	67.7	38.2	129.5	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	0	31	1
BTN	2007	275	40.1	25.3	50.8	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	0	0	0
MWI	2013	396	42.2	24.8	63.4	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0	0	0	0

Individual variables

The table below gives the type, observed range, and overall missingness of each analysis variable. It is a quick screen for variables that need closer review before modelling.

variable	class	missing_n	missing_prop	minimum	median	maximum
matmor	integer	426	0.115	2.000	66.000	2480.000

variable	class	missing_n	missing_prop	minimum	median	maximum
gdp_1000	numeric	62	0.017	0.110	4.072	123.679
infmor	numeric	20	0.005	1.600	18.900	138.100
male_edu	numeric	20	0.005	1.067	8.368	14.441
neomor	numeric	20	0.005	0.800	12.100	60.900
pop_dens	numeric	20	0.005	0.000	27.519	99.860
rainfall_1000	numeric	20	0.005	0.020	1.013	4.711
temp	numeric	20	0.005	-2.405	21.958	29.676
un5mor	numeric	20	0.005	2.000	22.200	224.900
urban	numeric	20	0.005	0.103	30.254	93.413
Drought	integer	0	0.000	0.000	0.000	1.000
Earthquake	integer	0	0.000	0.000	0.000	1.000
age_dep	numeric	0	0.000	16.172	55.512	111.477
conflict	integer	0	0.000	0.000	0.000	1.000
death_sum	integer	0	0.000	0.000	0.000	78644.000
oecd	integer	0	0.000	0.000	0.000	1.000

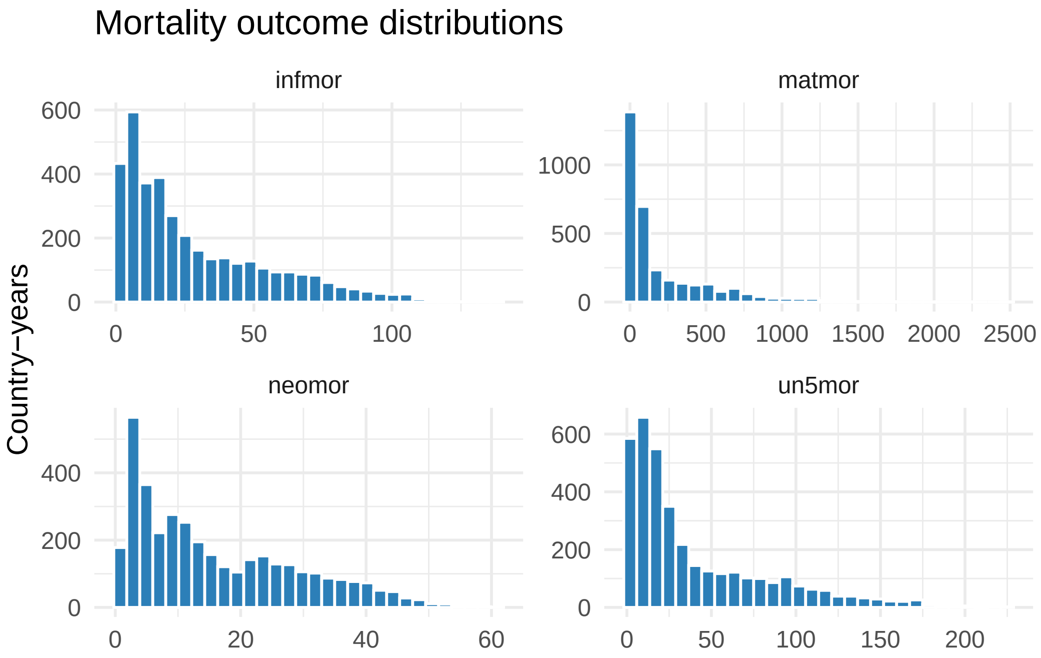

Figure 1: Distributions of mortality outcomes across country-years.

Missing data

The following plots show the proportion of missing values in each analysis variable. They help identify whether missingness is patterned by time, country, or conflict status; they do not establish a missing-data mechanism.

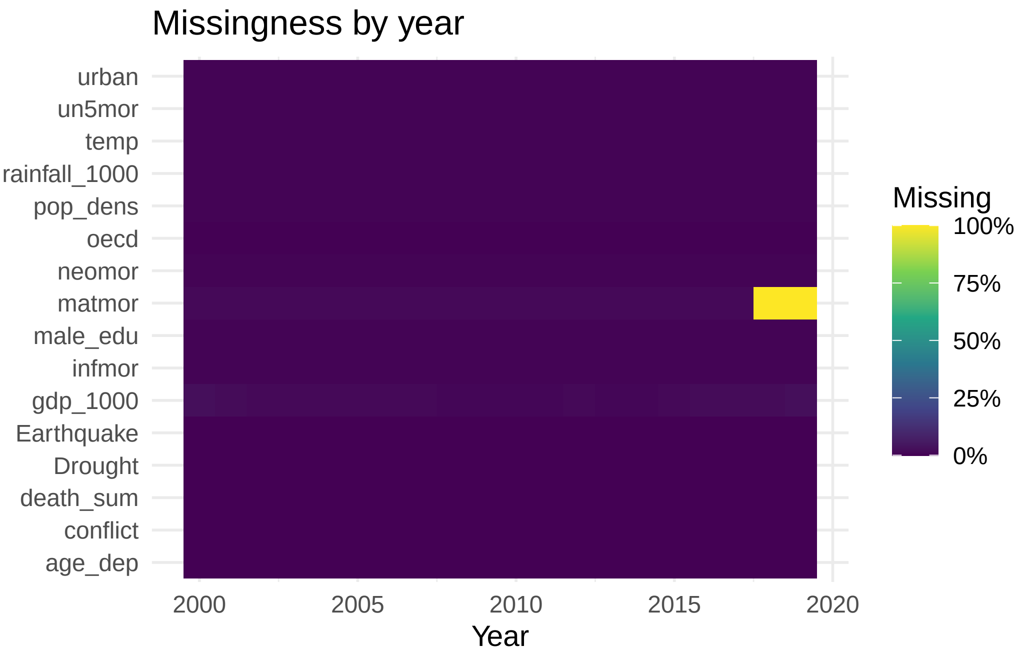

Figure 2: Proportion missing for each variable in each year.

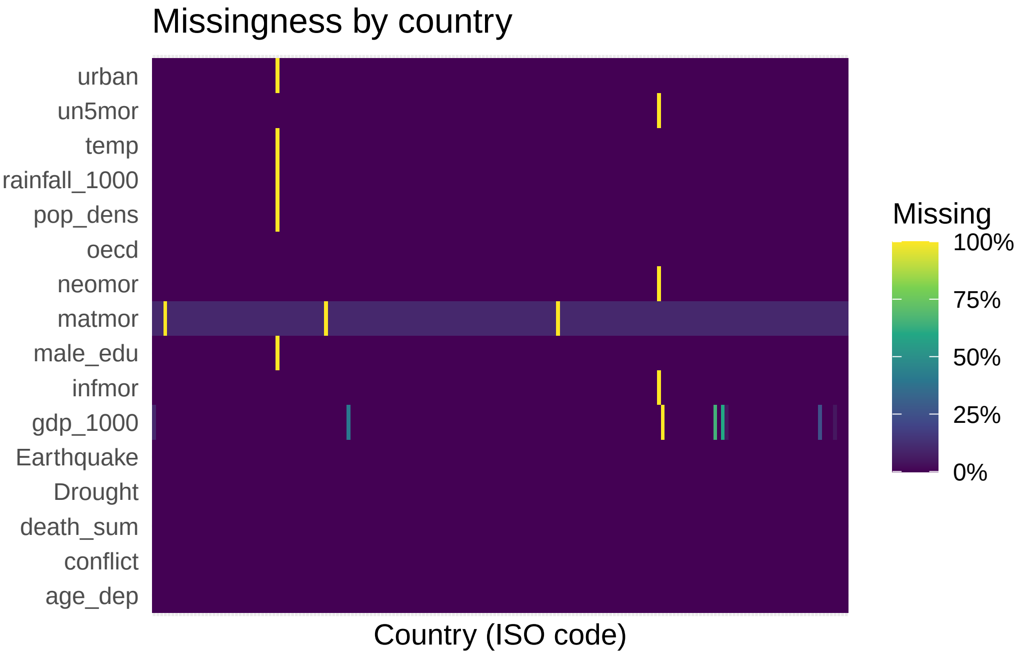

Figure 3: Proportion missing for each variable within each country over 2000–2019.

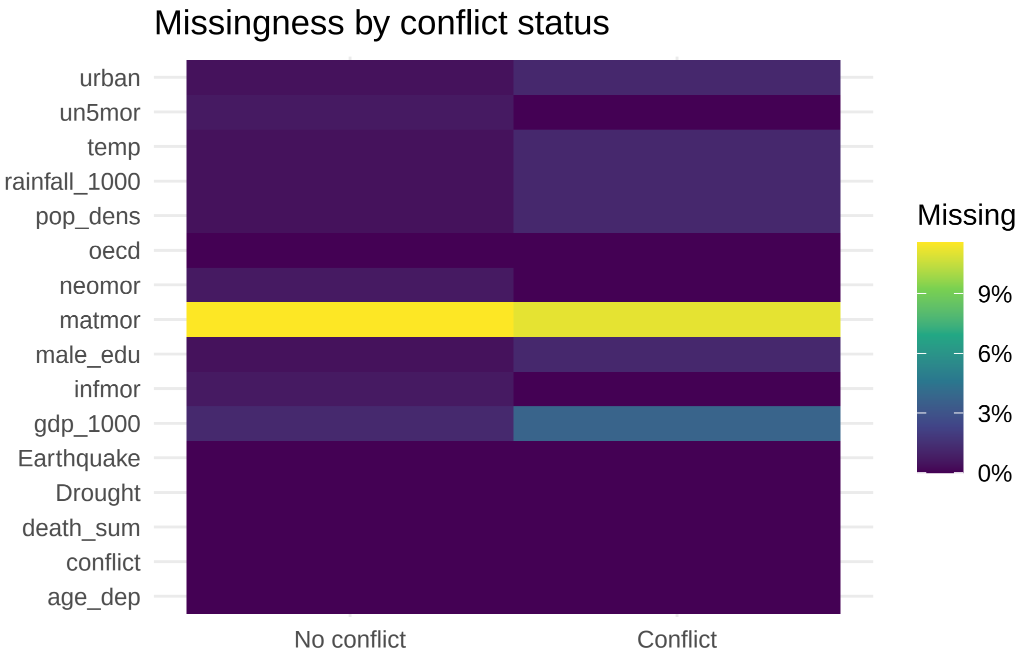

Figure 4: Proportion missing for each variable by conflict status.

Relationship between conflict and mortality outcomes

This descriptive plot compares mortality outcomes across the observed conflict indicator. It is not an adjusted or causal comparison.

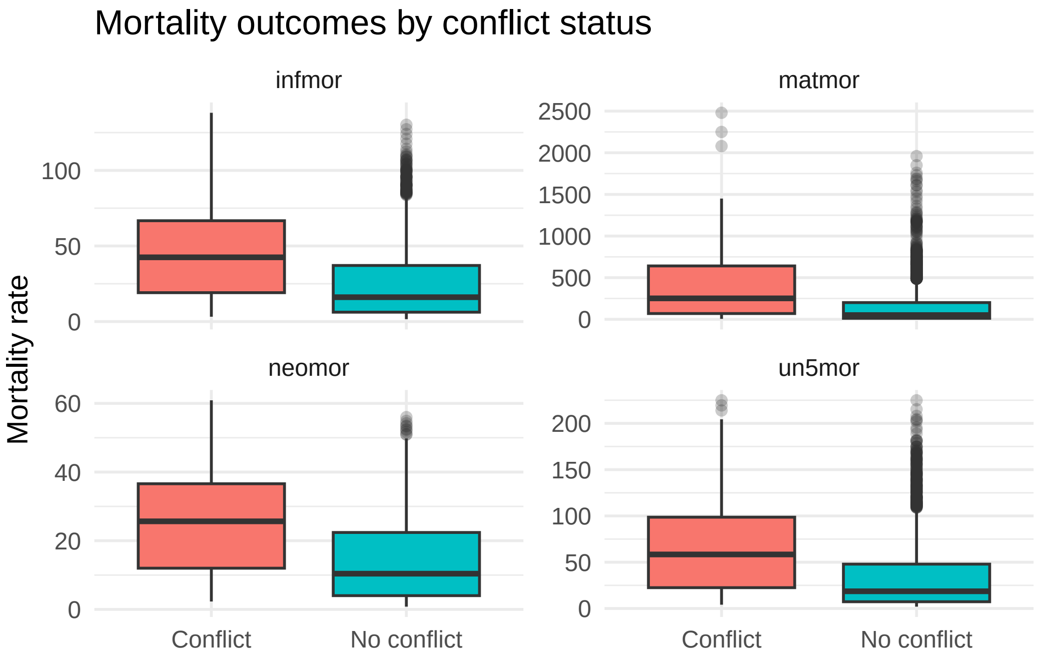

Figure 5: Mortality outcomes by conflict status.